BIRLA INSTITUTE OF TECHNOLOGY, MESRA, RANCHI

M.Sc. (Artificial Intelligence & Data Science)

CURRICULUM
BASED ON NATIONAL EDUCATION POLICY 2020
(Effective from Academic Session 2025-26)
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

BIRLA INSTITUTE OF TECHNOLOGY-MESRA, RANCHI
COURSE STRUCTURE FOR
MASTER OF SCIENCE (AI & DS)
(w.e.f. Academic Session 2025-26)
(Proposed)

SEMESTER 1: FIRST MONSOON (COURSE LEVEL: FOURTH)

Semester/Session of Study (Recommended)	Course Level	Category of Course	Courses	L	T	P	C
First Monsoon	Fourth	THEORY PC	SC25451 - Data Structures and Algorithms	3	0	0	3
		PC	SC25453 - Database Design Concepts	3	0	0	3
		PC	SC25455 - Programming for AI and DS	3	0	0	3
		PC	SC25457 - Operating Systems Concepts	3	0	0	3
		PC	SC25459 - Mathematics for AI & Data Science	3	0	0	3
		LABORATORIES PC	SC25452 - Data Structures and Algorithms Lab	0	0	3	1.5
		PC	SC25454 - Database Design Lab	0	0	3	1.5
		PC	SC25456 - AI and DS Programming Lab	0	0	3	1.5
		HSS	HS24131 - Communication Skills-I	0	0	3	1.5
TOTAL CREDITS							21

SEMESTER 2: SECOND SPRING (COURSE LEVEL: FOURTH - RECOMMENDED)

Semester/Session of Study (Recommended)	Course Level	Category of Course	Courses	L	T	P	C
Second Spring	Fourth	THEORY PC	SC25461 - Artificial Intelligence and Machine Learning	3	0	0	3
		PC	SC25463 - Design & Analysis of Algorithms	3	0	0	3
		PC	SC25465 - Data Science and Applications	3	0	0	3
		PE	PE - I (To be selected from Program Electives List)	3	0	0	3
		PE	PE - II (To be selected from Program Electives List)	3	0	0	3
		LABORATORIES PC	SC25462 - AI and ML Lab	0	0	3	1.5
		PC	SC25464 - Design & Analysis of Algorithms Lab	0	0	3	1.5
		PE	PE - I Lab (Related to PE-I course)	0	0	3	1.5
		HSS	HS24133 - Communication Skills-II	0	0	3	1.5
TOTAL CREDITS							21

PROGRAM ELECTIVES (PE)

Students need to select Program Electives (PE-I, PE-II, etc.) from the list below. These courses are offered in Semester 2 onwards based on availability and student interest.

Course Code	Category	Course Name	L	T	P	C
Note: Programme Electives are specialized courses that students can select from based on their interest and specialization area. The full list of electives and their availability across semesters will be provided by the department.						
Possible Areas	PE Categories	Examples (Not exhaustive)				
Machine Learning	PE	Advanced Machine Learning, Deep Learning, Reinforcement Learning, Natural Language Processing	3	0	0	3
Data Science	PE	Big Data Analytics, Data Mining, Time Series Analysis, Statistical Learning	3	0	0	3
AI Applications	PE	Computer Vision, Robotics, Autonomous Systems, Knowledge Graphs	3	0	0	3

Systems & Infrastructure	PE	Distributed Systems, Cloud Computing, GPU Computing, Edge AI	3	0	0	3

CURRICULUM OVERVIEW

Course Structure Summary:

• Programme Core (PC) Courses: Foundation courses in Data Structures, Algorithms, Database Design, AI, Machine Learning, and Data Science

• Programme Electives (PE): Specialized courses allowing students to choose based on their interests

• Humanities & Social Science (HSS): Communication Skills courses for overall development

• Laboratories: Hands-on practical experience with programming and AI/ML tools

Total Credits: 42 credits for 2-year M.Sc. programme (Semester 1: 21 credits, Semester 2: 21 credits)

Note: Additional semesters (Semester 3, Semester 4) may be included with project work, internships, and advanced electives depending on the full curriculum document.

COURSE CATEGORY ABBREVIATIONS

PC: Programme Core (compulsory courses covering fundamental concepts)

PE: Programme Elective (specialized courses for advanced topics)

HSS: Humanities and Social Science (soft skills and communication)

L: Lecture periods per week

T: Tutorial periods per week

P: Practical/Laboratory periods per week

C: Credits (Lecture: 1 period = 1 credit, Tutorial: 1 period = 0.5 credit, Practical: 1 period = 0.5 credit)

MODE OF DELIVERY:

L-Lecture; T-Tutorial; P-Practical/Laboratory

PROGRAMME STRUCTURE:

Duration: 2 years (4 semesters)

Total Credit Requirement: Minimum credits as per NEP 2020 guidelines

Assessment: Continuous evaluation including assignments, projects, seminars, and end-semester examinations